

A Career in Control and Automation

Involved in creating a safe, intelligent, power efficient and more productive environment, control and automation engineers engage with customers from multiple industry verticals. They also get immense exposure to a wide range of usage scenarios in the field of control and automation systems, which is a very good beginning to any career. Let's take a look at the scenario in India—the opportunities, scope and entry level jobs in control and automation



(Image courtesy: <http://www.career-cz6.jpg>)

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Machines are increasingly getting intelligent and this intelligence is provided by software. Currently, more functionalities are being defined by software rather than hardware. Sharing his point of view, Prasad Bhatt, vice president, Product Engineering Solutions, Wipro, says that, "We observe that IT has a very important role to play on the shop floor, and to drive this trend, the industry needs automation engineers who understand the interplay of software and machines."

Starting from waste and water management to any production assembly line, there is a need for control and automation systems, believes Partamita Kapat, head-HR, elfnchips.

She adds, "Synchronised signals and correct signal indications can create an efficient system to make life easier. Intelligence can be added to such systems by 'monitoring' the traffic and flow."

It's a multi-disciplinary field

The control and automation domain is a multi-disciplinary field. It requires a convergence of various engineering fields—CS/IT, electrical, mechanical, chemical and electronics to develop a solution. "Control and automation requires a blend of process knowledge, logic theory, software/platform knowledge, instrumentation and IT," says Amit Kumar Kushawaha, product manager (EMS-Energy Management Systems), Rishabh Instruments Pvt Ltd, Nashik.

The basic premise of automation is to pass on the intelligence to a system with the objective of making it efficient and economical. "Systems such as energy management are an investment and lead to savings in aspects of consumption, man hours and enhanced system availability," Kushawaha adds.

The control and automation domain is currently in a situation in which demand far outstrips supply. "In fact, this is one of the few technical areas in the electrical/electronic-engineering field where the knowledge or skills available in India are seriously inadequate, and students, engineers and experts in these areas will be required by the industry in the immediate future," says Nilesh Sawant, head, SITRAIN India, Siemens Ltd. Prasad Bhatt believes that in India, the adoption of automation has lagged behind countries in Europe and North America. He adds, "However, some companies, especially those in process industries such as steel, cement and oil & gas, have been leading adopters of automation technologies."

The Indian market for automation is evolving, believes Kushawaha. With the rising cost of resources, stringent regulations, and the increasing concern for the environment and safety, companies are making systems efficient. But within the current economic scenario, the decision to switch to automation is also based on concerns about cash flow and capex. There is also the need to raise awareness among end customers. Kushawaha says, "Automation has percolated to several new areas including security, lighting, fire detection, entertainment and energy